



Touch The Future Together With Multi-Touch Technology

We not only see touch technology in science fiction movies, but also in real life across thousands of businesses that focus on customer engagement



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Tim Berners-Lee, inventor of World Wide Web, once said, "There was a time when people felt the Internet was another world, but now people realise it is a tool we use in this world."

Technological progress allows more and more inventions to occur that open a new world of possibilities for all types of businesses, creating a link with people around the world to make connections easily and, hence, developing strong bond-age between them. Electronic devices like cellphones, trackpads and laptops have become ingrained in our daily lives.

The physical keyboard in these devices was replaced by the sleek touchscreen. This technology was initiated with the capacitive touchscreen (used in air traffic control, and not by people) invented by E.A. Johnson around 1965-1967, at Royal Radar Establishment, UK, followed by the resistive touchscreen invented by George Samuel Hurst in 1982. This was followed by Hewlett-Packard's built-in infrared (IR) beams that detected finger movements. Later, Apple introduced its Newton personal digital assistant (PDA)

with handwriting recognition in the 1990s, while IBM released Simon, the first smartphone with touchscreen interface. In 1996, Palm released its Pilot Series with advanced touchscreen technology. In 2002, Microsoft entered the touch technology sphere by introducing Windows XP Tablet edition, and in 2007, Apple introduced its iPhone, the king of smartphones with touchscreen.

Single user versus multi-users

A single user using single (for example, ATM) or dual touch on a smartphone, like selecting an icon, typing a letter or using two fingers to zoom in or out, is an attractive digital trend compared to the traditional keypad phone. This classic touchscreen has two limitations: only a single user is allowed at a time due to single input command and it is limited based on scale, that is, cost of the touchscreen increases as size of the screen increases.

As human needs are wide, the single-user single-touch did not offer much in terms of flexibility and expansion into more command inputs. This led to the invention of multi-touch user touchscreens. We already

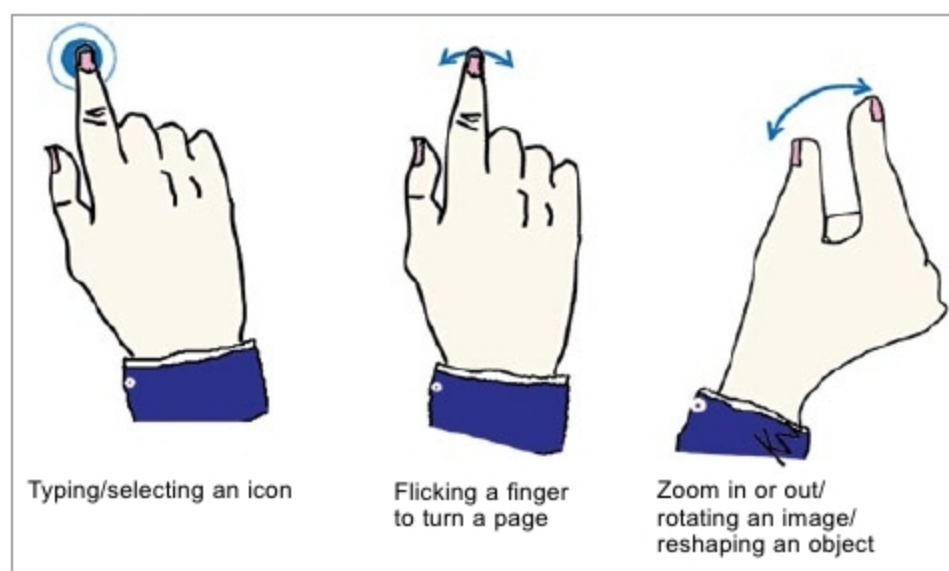


Fig. 1: Multi-touching in smartphones or tablets

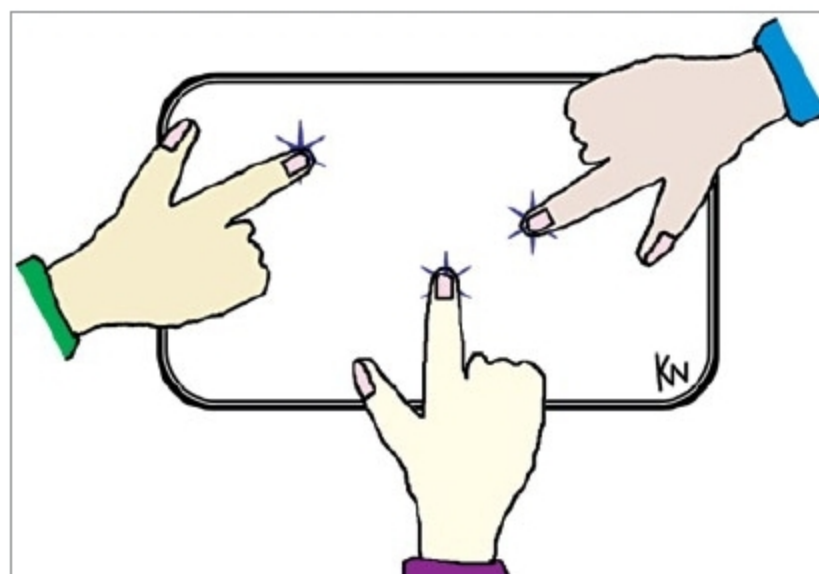


Fig. 2: Multi-touch technology